

1. A mixture of O₂ and N₂ with their partial pressures in the ratio 0.21 to 0.79 is in a container at 25°C. Calculate the molar concentration, the mass density, and the mass fraction of each species for a total pressure of 1 bar. What would be the average molecular weight of the mixture?

Given:

Partial pressure of O₂, P_{O₂} = 0.21 × Total pressure

$$= 0.21 \times 1 \text{ bar}$$

$$= 0.21 \times 1 \times 10^5 \text{ N/m}^2$$

Partial pressure of N₂, P_{N₂} = 0.79 × Total pressure

$$= 0.79 \times 1 \text{ bar}$$

$$= 0.79 \times 1 \times 10^5 \text{ N/m}^2$$

Temperature, T = 25°C + 273

$$= 298 \text{ K}$$

To find:

1. Molar concentrations, C_{O₂}, C_{N₂}
2. Mass densities, ρ_{O₂}, ρ_{N₂}
3. Mass fractions, m_{O₂}, m_{N₂}
4. Average molecular weight, M

Solution:

We know that,

$$\text{Molar concentration, } C = \frac{P}{RT}$$

$$\begin{aligned} \Rightarrow C_{O_2} &= \frac{P_{O_2}}{RT} \\ &= \frac{0.21 \times 1 \times 10^5}{8314 \times 298} \end{aligned}$$

[∵ Universal gas constant, R = 8314 J/kg – mole – K]

$C_{O_2} = 8.476 \times 10^{-3} \text{ kg – mole /m}^3$

$$\begin{aligned} \Rightarrow C_{N_2} &= \frac{P_{N_2}}{RT} \\ &= \frac{0.79 \times 1 \times 10^5}{8314 \times 298} \end{aligned}$$

$$C_{N_2} = 31.88 \times 10^{-3} \text{ kg - mole /m}^3$$

We know that,

$$\text{Molar concentration, } C = \frac{P}{M}$$

$$\begin{aligned} \Rightarrow \rho_{O_2} &= C_{O_2} \times M_{O_2} \\ &= 8.476 \times 10^{-3} \times 32 \end{aligned}$$

[$\because$ Molecular weight of O_2 is 32]

$$\rho_{O_2} = 0.271 \text{ kg/m}^3$$

$$\begin{aligned} \Rightarrow \rho_{N_2} &= C_{N_2} \times M_{N_2} \\ &= 31.88 \times 10^{-3} \times 28 \end{aligned}$$

[$\because$ Molecular weight of N_2 is 28]

$$\rho_{N_2} = 0.893 \text{ kg/m}^3$$

Overall density, $\rho = \rho_{O_2} + \rho_{N_2}$

$$= 0.271 + 0.893$$

$$\rho = 1.164 \text{ kg / m}^3$$

Mass fractions:

$$m_{O_2} = \frac{\rho_{O_2}}{\rho} = \frac{0.271}{1.164}$$

$$m_{O_2} = 0.233$$

$$m_{N_2} = \frac{\rho_{N_2}}{\rho} = \frac{0.893}{1.164}$$

$m_{N_2} = 0.767$

Average Molecular weight,

$$\begin{aligned} M &= P_{O_2} M_{O_2} + P_{N_2} M_{N_2} \\ &= 0.21 \times 32 + 0.79 \times 28 \end{aligned}$$

$M = 28.84$

Result:

1. $C_{O_2} = 8.476 \times 10^{-3} \text{ kg - mole / m}^3$
 $C_{N_2} = 31.88 \times 10^{-3} \text{ kg - mole / m}^3$
2. $\rho_{O_2} = 0.271 \text{ kg / m}^3$
 $\rho_{N_2} = 0.893 \text{ kg / m}^3$
3. $m_{O_2} = 0.233$
 $m_{N_2} = 0.767$
4. $M = 28.84$

2. The molecular weights of the two components A and B of a gas mixture are 24 and 28 respectively. The molecular weight of a gas mixture is found to be 30. If the mass concentration of the mixture is 1.2 kg/m^3 , determine the following:

- I. Density of component A and B**
- II. Molar fractions**
- III. Mass fractions**
- IV. Total pressure if the temperature of the mixture is 290 K**

Given:

Molecular weight of component A, $M_A = 24$

Molecular weight of component B, $M_B = 48$

Molecular weight of gas mixture, $M = 30$

Mass concentration, $\rho = 1.2 \text{ kg/m}^3$

Temperature, $T = 290 \text{ K}$

To find:

1. Density of component A and B, ρ_A, ρ_B
2. Molar fraction, x_A , and x_B
3. Mass fractions, m_A , and m_B
4. Total pressure, P

Solution:

Molar concentration of the mixture,

$$C = \frac{\rho}{M}$$
$$= \frac{1.2}{30}$$

$$C = 0.04$$

We know that,

$$C_A + C_B = C$$

$$C_A + C_B = 0.04$$

..... (1)

We know that,

$$\rho_A = M_A C_A$$

$$[\because M_A = 24]$$

$\Rightarrow$

$$\rho_A = 24 C_A$$

$$\rho_B = M_B C_B$$

$$[\because M_B = 48]$$

$\Rightarrow$

$$\rho_B = 48 C_B$$

We know that,

$$\rho_A + \rho_B = \rho$$

$\Rightarrow$

$$24 C_A + 48 C_B = \rho$$

..... (2)

Solving equation (1) and (2)

$$\Rightarrow C_A = 0.03 \text{ kg mole/m}^3$$

$$\Rightarrow C_B = 0.01 \text{ kg mole/m}^3$$

(i) Density

$$\text{Density, } \rho_A = 24C_A$$

$$= 24 \times 0.03$$

$$\rho_A = 0.72 \text{ kg / m}^3$$

$$\text{Density, } \rho_B = 48C_A$$

$$= 24 \times 0.01$$

$$\rho_B = 0.48 \text{ kg / m}^3$$

ii) Mole fractions

$$x_A = \frac{C_A}{C} = \frac{0.03}{0.04} = 0.75$$

$$x_B = \frac{C_B}{C} = \frac{0.01}{0.04} = 0.25$$

iii) Mass fractions

$$m_A = \frac{\rho_A}{\rho} = \frac{0.72}{1.2} = 0.06$$

$$m_B = \frac{\rho_B}{\rho} = \frac{0.48}{1.2} = 0.4$$

iv) Total pressure at 290 K

Gas law, $pV = mRT$

$$\Rightarrow P = \frac{m}{V}RT$$

$$= \rho RT \quad \left[\because R = \frac{G}{M} \right]$$

$$= \rho \times \frac{G}{M} \times T$$

$$= 1.2 \times \frac{8314}{30} \times 290$$

$$\rho_B = 48 \text{ kg} / \text{m}^3$$

[$\because$ Universal gas constant, $G = 8314 \text{ J/kg} - \text{mole} - \text{K}$]

$$P = 96442 \text{ N/m}^2$$

$P = 96.442 \text{ kN/m}^2$

Result:

1. $\rho_A = 0.72 \text{ kg} / \text{m}^3$
 $\rho_B = 0.48 \text{ kg} / \text{m}^3$
2. $x_A = 0.75$
 $x_B = 0.25$
3. $m_A = 0.6$
 $m_B = 0.4$
4. $P = 96.442 \text{ kN} / \text{m}^2$

3. Gaseous hydrogen is stored in a rectangular container. The walls of the container are of steel having 25 mm thickness. At the inner surface of the container, the molar concentration of hydrogen in the steel is 1.2 kg mole/m^3 while at the outer surface of the container the molar concentration is zero. Calculate the molar diffusion flux for hydrogen through the steel. Take diffusion co-efficient for hydrogen in steel is $0.24 \times 10^{-12} \text{ m}^2/\text{s}$

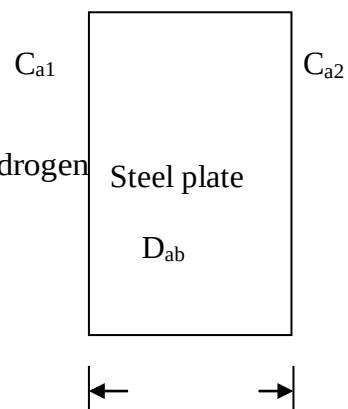
Given:

Thickness, $L = 2 \text{ mm} = 0.002 \text{ m}$

Molar concentration at inner side, $C_{a1} = 1.2 \frac{\text{kg} - \text{mole}}{\text{m}^3}$ Hydrogen

Molar concentration at outer side,

$$C_{a2} = 0$$



Diffusion co-efficient,

$$D_{ab} = 0.24 \times 10^{-12} \text{ m}^2/\text{s}$$

To find:

$$\text{Molar diffusion flux, } \frac{m_a}{A}$$

Solution:

We know that, for plane membrane

$$\begin{aligned} \text{Molar flux, } \frac{m_a}{A} &= \frac{D_{ab}}{A} [C_{a1} - C_{a2}] \\ &= \frac{0.24 \times 10^{-12}}{0.25} [1.2 - 0] \\ \frac{m_a}{A} &= 1.15 \times 10^{-11} \frac{\text{kg-mole}}{\text{s-m}^2} \end{aligned}$$

Result:

$$\text{Molar diffusion flux, } \frac{m_a}{A} = 1.15 \times 10^{-11} \frac{\text{kg-mole}}{\text{s-m}^2}$$

4. CO₂ and air experience equimolar counter diffusion in a circular tube whose length and diameter are 1.2 m and 60 mm respectively. The system is at a total pressure of 1 atm and a temperature of 273 K. The ends of the tube are connected to large chambers. Partial pressure of CO₂ at one end is 200 mm of Hg while at the other end is 90 mm of Hg. Calculate the following

- 1. Mass transfer rate of CO₂ and**
- 2. Mass transfer rate of air**

Given:

Diameter, $d = 60 \text{ mm} = 0.06 \text{ m}$

Length, $(x_2 - x_1) = 1.2 \text{ m}$

Total pressure, $p = 1 \text{ atm} = 1 \text{ bar}$

Temperature, $T = 273 \text{ K}$

Partial pressure of CO₂ at one end

$$P_{a1} = 200 \text{ mm of Hg} = \frac{90}{760} \text{ bar}$$

$$\Rightarrow P_{a1} = 0.263 \text{ bar} \quad [\because 1 \text{ bar} = 760 \text{ mm of Hg}]$$

$$\Rightarrow \boxed{P_{a1} = 0.263 \times 10^5 \text{ N/m}^2}$$

$$[\because 1 \text{ bar} = 10^5 \text{ N/m}^2]$$

Partial pressure of CO₂ at other end

$$P_{a2} = 90 \text{ mm of Hg} = \frac{90}{760} \text{ bar}$$

$$\Rightarrow P_{a2} = 0.118 \text{ bar}$$

$$\Rightarrow P_{a2} = 0.118 \times 10^5 \text{ N/m}^2$$

To find:

1. Mass transfer rate of CO₂
2. Mass transfer rate of air

Solution:

We know that, for equimolar diffusion

$$\text{Molar flux, } \frac{m_a}{A} = \frac{D_{ab}}{GT} \left[\frac{P_{a1} - P_{a2}}{x_2 - x_1} \right] \quad \dots\dots\dots (1)$$

Where,

D_{ab} – diffusion co-efficient – m²/s

The diffusion co-efficient for CO₂ – Air combination is $11.86 \times 10^{-6} \text{ m}^2/\text{s}$

$$D_{ab} = 11.89 \times 10^{-6} \text{ m}^2/\text{s}$$

$$G = \text{Universal gas constant} = 8314 \frac{\text{J}}{\text{kg} - \text{mole} - \text{K}}$$

$$A - \text{Area} = \frac{\pi}{4} d^2$$

$$\frac{\pi}{4} (0.060)^2$$

$$A = 2.82 \times 10^{-3} \text{ m}^2$$

$$(1) \Rightarrow \frac{m_a}{2.82 \times 10^{-3}} = \frac{11.89 \times 10^{-6}}{83.14 \times 273} \times \left[\frac{0.263 \times 10^5 - 0.118 \times 10^5}{1.2} \right]$$

Molar transfer rate of CO ₂ , $m_a = 1.785 \times 10^{-10} \frac{\text{kg} - \text{mole}}{\text{s}}$

We know,

Mass transfer rate of CO₂ = Molar transfer × Molecular weight

$$= 1.785 \times 10^{-10} \times 44.01$$

[Molecular weight of CO₂ = 44.01, refer HMT data, page no. 182 0]

$$\text{Mass transfer rate of air, } m_b = - 1.785 \times 10^{-10} \frac{\text{kg} - \text{mole}}{\text{s}} \quad [\because m_a = - m_b]$$

Mass transfer rate of air = Molar transfer × Molecular weight or air

$$= - 1.785 \times 10^{-10} \times 29$$

Mass transfer rate of air = $- 5.176 \times 10^{-9} \text{ kg/s}$

Result:

1. Mass transfer rate of CO₂ = $7.85 \times 10^{-9} \text{ kg/s}$
2. Mass transfer rate of air = $- 5.176 \times 10^{-9} \text{ kg/s}$

5. Determine the diffusion rate of water from the bottom of a test tube of 25 mm diameter and 35 mm long into dry air at 25°C. Take diffusion co-efficient of water in air is $0.28 \times 10^{-4} \text{ m}^2/\text{s}$.

Given:

Diameter, $d = 25 \text{ mm} = 0.025 \text{ m}$

Length, $(x_2 - x_1) = 35 \text{ mm} = 0.035 \text{ m}$

Temperature, $T = 25^\circ\text{C} + 273 = 298 \text{ K}$

Diffusion co-efficient, $D_{ah} = 0.28 \times 10^{-4} \text{ m}^2/\text{s}$

To find:

Diffusion rate of water

Solution:

We know that, for isothermal evaporation,

$$\text{Molar flux, } \frac{m_a}{A} = \frac{D_{ab}}{GT} \left(\frac{P}{x_2 - x_1} \right) \ln \left(\frac{p - p_{w2}}{p - p_{w1}} \right) \quad \dots\dots\dots (1)$$

[From equation no. 5.9]

Where,

$$G - \text{Universal gas constant} = 8314 \frac{J}{kg - mole - K}$$

$$p - \text{Total pressure} = 1 \text{ atm} = 1.013 \text{ bar} = 1.013 \times 10^5 \text{ N/m}^2$$

p_{w1} = Partial pressure at the bottom of the test tube corresponding to saturation temperature 25°C

At 25°C

$$\Rightarrow P_{w1} = 0.03166 \text{ bar}$$

$$\Rightarrow \boxed{P_{w1} = 0.03166 \times 10^5 / \text{m}^2}$$

P_{w2} – Partial pressure at the top of the test tube. Here, air is dry and there is no water vapour. So, $p_{w2} = 0$.

$$\Rightarrow \boxed{p_{w2} = 0}$$

$$\text{Area, } A = \frac{\pi d^2}{4}$$

$$= \frac{\pi (0.025)^2}{4}$$

$$\boxed{A = 4.90 \times 10^{-4} \text{ m}^2}$$

$$(1) \Rightarrow \frac{ma}{4.90 \times 10^{-4}} = \frac{0.28 \times 10^{-4}}{8314 \times 298} \times \frac{1.013 \times 10^5}{0.035} \times \ln \left(\frac{1.013 \times 10^5 - 0}{1.013 \times 10^5 - 0.03166 \times 10^5} \right)$$

$$\text{Molar rate of water vapour, } m_a = 5.09 \times 10^{-10} \frac{\text{kg} - \text{mole}}{\text{s}}$$

We know that,

$$\begin{aligned} \text{Mass rate of water vapour} &= \text{molar rate of water vapour} \times \text{Molecular weight of water vapour} \\ &= 5.09 \times 10^{-10} \times 18.016 \end{aligned}$$

[$\because$ Molecular weight of steam = 18.016, refer HMT data book, page no. 183]

$$\text{Mass transfer rate of water vapour} = 9.170 \times 10^{-9} \text{ kg / s}$$

Result:

$$\text{Diffusion rate of water} = 9.170 \times 10^{-9} \text{ kg/s}$$

6. Dry air at 30°C and one atmospheric pressure flows over a flat plate of 600 mm long at a velocity of 55 m/s. Calculate the mass transfer co-efficient at the plate.

Given:

Fluid temperature, $T_{\infty} = 30^{\circ}\text{C}$

Velocity, $U = 55 \text{ m/s}$

Length, $x = 600 \text{ mm} = 0.6 \text{ m}$

To find:

Mass transfer co-efficient, (h_m)

Solution:

Properties of air at 30°C

Kinematic viscosity, $\nu = 16 \times 10^{-6} \text{ m}^2/\text{s}$

We know that,

$$\begin{aligned} \text{Reynolds Number, } Re &= \frac{Ux}{\nu} \\ &= \frac{55 \times 0.6}{16 \times 10^{-6}} \\ Re &= 2.06 \times 10^6 > 5 \times 10^5 \end{aligned}$$

Since, $Re > 5 \times 10^5$, flow is turbulent

[Flow is laminar upto $Re = 5 \times 10^5$, after that flow is turbulent]

For combined Laminar – Turbulent flow, flat plate,

$$\text{Sherwood Number (Sh)} = [0.037 (Sh)^{0.8} - 871] Sc^{0.333} \dots \dots \dots (1)$$

Where,

$$Sc - \text{Schmidt Number} = \frac{\nu}{D_{ab}} \dots \dots \dots (2)$$

D_{ab} – Diffusion co-efficient (water + Air) at $30^{\circ}\text{C} \approx 26^{\circ}\text{C} = 25.83 \times 10^{-6} \text{ m}^2/\text{s}$

$D_{ab} = 25.83 \times 10^{-6} \text{ m}^2/\text{s}$
--

$$(2) \Rightarrow Sc = \frac{16 \times 10^{-6}}{25.83 \times 10^{-6}}$$

$$Sc = 0.619$$

Substitute Sc, Re values in equation (1)

$$(1) \Rightarrow sh = [0.037 (2.06 \times 10^6)^{0.8} - 871] (0.619)^{0.333}$$

$$Sh = 2805.13$$

We know that,

$$\text{Sherwood Number, } Sh = \frac{h_m x}{D_{ab}}$$

$$\Rightarrow 2805.13 = \frac{h_m \times 0.6}{25.83 \times 10^{-6}}$$

$$\text{Mass transfer co-efficient, } h_m = 0.121 \text{ m/s}$$

Result:

$$\text{Mass transfer co-efficient, } h_m = 0.121 \text{ m/s}$$

7. Air at 25°C flows over a tray full of water with a velocity of 2.8 m/s. The tray measures 30 cm along the flow direction and 40 cm wide. The partial pressure of water present in the air is 0.007 bar. Calculate the evaporation rate of water if the temperature on the water surface is 15°C. Take diffusion co-efficient is $4.2 \times 10^{-5} \text{ m}^2/\text{s}$.

Given:

$$\text{Fluid temperature, } T_{\infty} = 25^{\circ}\text{C}$$

$$\text{Speed, } U = 2.8 \text{ m/s}$$

$$\text{Flow direction is 30 cm side. So, } x = 30 \text{ cm} = 0.30 \text{ m}$$

$$\text{Area, } A = 30 \text{ cm} \times 40 \text{ cm} = 0.30 \times 0.40 \text{ m}^2$$

$$\text{Partial pressure of water, } p_{w2} = 0.007 \text{ bar}$$

$$P_{w2} = 0.007 \times 10^{-5} \text{ N/m}^2$$

Water surface temperature, $T_w = 15^\circ\text{C}$

Diffusion co-efficient, $D_{ab} = 4.2 \times 10^{-5} \text{ N/m}^2$

To find:

Evaporation rate of water, (m_w)

Solution:

We know that,

$$\text{Film temperature, } T_f = \frac{T_w + T_\infty}{2} = \frac{15 + 25}{2}$$

$$T_f = 20^\circ\text{C}$$

Properties of air at 20°C

Kinematic viscosity, $\nu = 15.6 \times 10^{-6} \text{ m}^2/\text{s}$

We know that,

$$\begin{aligned} \text{Reynolds Number, } Re &= \frac{Ux}{\nu} \\ &= \frac{2.8 \times 0.30}{15.06 \times 10^{-6}} \\ Re &= 0.557 \times 10^5 < 5 \times 10^5 \end{aligned}$$

Since, $Re < 5 \times 10^5$, flow is laminar.

For flat plate, Laminar flow:

$$\text{Sherwood Number, (Sh)} = [0.664 (Re)^{0.5} (Sc)^{0.333}] \dots\dots\dots (1)$$

Where,

$$Sc - \text{Schmidt Number} = \frac{\nu}{D_{ab}} \dots\dots\dots (2)$$

$$Sc = \frac{15.06 \times 10^{-6}}{4.2 \times 10^{-5}}$$

$$Sc = 0.358$$

Substitute Sc, Re values in equation (1)

$$(1) \rightarrow sh = [0.664 (0.557 \times 10^5)^{0.5} (0.358)^{0.333}]$$

$$Sh = 111.37$$

We know that,

$$\text{Sherwood Number, } Sh = \frac{h_m x}{D_{ab}}$$

$$\Rightarrow 111.37 = \frac{h_m \times 0.30}{4.2 \times 10^{-5}}$$

$$\text{Mass transfer co-efficient, } h_m = 0.0155 \text{ m/s}$$

Mass transfer co-efficient based on pressure difference is given by,

$$h_{mp} = \frac{h_m}{RT_w} = \frac{0.0155}{287 \times 288}$$

$$[\because T_w = 15^\circ\text{C} + 273 = 288 \text{ K, } R = 287 \text{ J/Kg K}]$$

$$h_{mp} = 1.88 \times 10^{-7} \text{ m/s}$$

Saturation pressure of water at 15°C

$$P_{w1} = 0.017$$

$$P_{w1} = 0.017 \times 10^5 \text{ N/m}^2$$

The evaporation rate of water is given by,

$$\begin{aligned} m_w &= h_{mp} \times A [p_{w1} - p_{w2}] \\ &= 1.88 \times 10^{-7} \times (0.30 \times 0.40) \times [0.017 \times 10^5 - 0.007 \times 10^5] \end{aligned}$$

$$m_w = 2.25 \times 10^{-5} \text{ kg/s}$$

Result:

Evaporation rate of water, $m_w = 2.25 \times 10^{-5} \text{ kg/s}$

8. Air at 30°C and atmospheric pressure flows in a 12 mm diameter tube of 1 meter length with a velocity of 2.5 m/s. The inside surface of the tube contains a deposit of naphthalene.

Determine the average mass transfer co-efficient. Take diffusion co-efficient, $D_{ab} = 0.62 \times 10^{-5} \text{ m}^2/\text{s}$.

Give:-

Fluid temperature, $T_\infty = 30^\circ\text{C}$

Velocity, $U = 2.5 \text{ m/s}$

Diameter, $D = 12 \text{ mm} = 0.012 \text{ m}$

Length, $x = 1 \text{ m}$

Diffusion co-efficient, $D_{ab} = 0.62 \times 10^{-5} \text{ m}^2/\text{s}$

To find:

Average mass transfer co-efficient, h_m

Solution:

Properties of air at 30°C

Kinematic viscosity, $\nu = 16 \times 10^{-6} \text{ m}^2/\text{s}$

We know that,

$$\begin{aligned} \text{Reynolds Number, } Re &= \frac{UD}{\nu} \\ &= \frac{2.5 \times 0.012}{16 \times 10^{-6}} \\ Re &= 1875 \times 2000 \end{aligned}$$

Since, $Re < 2000$, flow is laminar.

For laminar internal flow:

Sherwood Number, $Sh = 3.66$

We know that,

$$\begin{aligned} Sh &= \frac{h_m D}{D_{ab}} \\ \Rightarrow 3.66 &= \frac{h_m \times 0.012}{D_{ab}} \\ \Rightarrow 3.66 &= \frac{h_m \times 0.012}{0.62 \times 10^{-5}} \end{aligned}$$

Mass transfer co-efficient, $h_m = 1.89 \times 10^{-3}$ m/s.

Result:

Mass transfer co-efficient, $h_m = 1.89 \times 10^{-3}$ m/s